

# Does Contract Commitment Reduce Customer Churn?

A causal analysis of the IBM Telco customer-churn dataset, produced end-to-end with the aitiolin Causality Agent. Revised edition with a real sensitivity analysis and a reviewed set of caveats.

**Dataset:** Churn.csv (IBM Telco; 21 raw variables, 7,043 customers)

**Treatment:** Contract (month-to-month → one-year → two-year, coded 0/1/2)

**Outcome:** Churn (customer left within the observation window, 0/1; base rate 26.5%)

**Adjustment set identified:** gender, tenure

**Model source:** two LLM-assisted agent drafts, statistically verified, merged by the analyst

## Headline result

Moving a customer one contract tier up is associated with **14.5% lower churn probability** after adjusting for tenure (the classic confounder: long-standing customers both choose longer contracts and churn less anyway). The naive comparison (-0.227) overstates the effect by ~56%; even the deliberately hostile confounder-stressed model keeps 4.8%. The estimate has  $t = -19.9$ , survives every refutation test, and would need an unobserved confounder explaining about 21% of the residual variance of both contract choice and churn to be explained away - stronger on the outcome side than any variable in the dataset. Contract commitment is a genuine, actionable lever - with the important caveat that part of “commitment” is contractual lock-in rather than earned loyalty.

## 1. The question and the data

Month-to-month customers churn far more (42.7%) than one-year (11.3%) or two-year (2.8%) customers. Does the contract *cause* retention, or do loyal, long-tenure customers self-select into longer contracts? Average tenure is 18 months on month-to-month, 42 on one-year and 57 on two-year plans - tenure is the obvious confounder and it is large.

Data: IBM Telco customer churn, 7,043 customers x 21 columns - demographics, subscribed services, account attributes (tenure, contract, billing, payment method, charges) and the churn label.

### Data preparation (Causality Agent, stages 1-2)

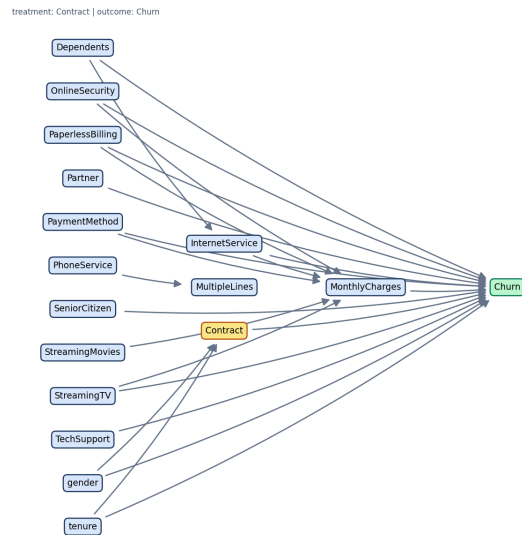
| Column       | Action         | Result                                                  |
|--------------|----------------|---------------------------------------------------------|
| customerID   | drop column    | Dropped column 'customerID'.                            |
| TotalCharges | impute missing | Imputed 11 value(s) in 'TotalCharges' with median 1397. |

## 2. Building the causal model: two drafts and a merge

The agent's **first draft** (12 edges) pointed every edge straight into Churn - a “star” model with no confounding paths, i.e. the naive feature-importance view. Re-prompting for structure *among* covariates produced a **second draft** (15 edges: tenure → Contract, InternetService → MonthlyCharges, ...). The analyst merge took the union of all non-rejected edges of both drafts and dropped the lowest-confidence edge on any cycle. The merged graph:

| Edge                              | Verifier verdict | Confidence |
|-----------------------------------|------------------|------------|
| Contract -> Churn                 | supported        | 1.00       |
| TechSupport -> Churn              | supported        | 1.00       |
| OnlineSecurity -> Churn           | supported        | 1.00       |
| MonthlyCharges -> Churn           | supported        | 0.83       |
| PaperlessBilling -> Churn         | supported        | 0.83       |
| Dependents -> Churn               | weak             | 0.70       |
| tenure -> Churn                   | supported        | 1.00       |
| InternetService -> Churn          | conflict         | 0.50       |
| PaymentMethod -> Churn            | weak             | 0.70       |
| StreamingTV -> Churn              | conflict         | 0.50       |
| gender -> Churn                   | conflict         | 0.50       |
| SeniorCitizen -> Churn            | supported        | 0.83       |
| tenure -> Contract                | supported        | 1.00       |
| InternetService -> MonthlyCharges | supported        | 1.00       |
| gender -> Contract                | conflict         | 0.50       |
| Dependents -> InternetService     | conflict         | 0.50       |

| Edge                               | Verifier verdict | Confidence |
|------------------------------------|------------------|------------|
| PhoneService -> MultipleLines      | conflict         | 0.50       |
| OnlineSecurity -> MonthlyCharges   | conflict         | 0.50       |
| PaperlessBilling -> MonthlyCharges | supported        | 1.00       |
| PaymentMethod -> MonthlyCharges    | supported        | 0.83       |
| StreamingTV -> MonthlyCharges      | supported        | 1.00       |
| StreamingMovies -> MonthlyCharges  | supported        | 1.00       |
| Partner -> Churn                   | conflict         | 0.62       |



Rendered causal graph as saved to the canvas.

*Agent's reasoning (draft 2):* This causal structure is plausible as it captures the complex relationships between customer demographics, service choices, and churn behavior. The outcome of interest, Churn, is influenced by various factors, including the type of Contract, which can be seen as an actionable lever for retention strategies. The edges represent mechanisms such as how tenure influences contract choices and how different service types affect monthly charges, which in turn can impact customer satisfaction and retention. The biggest structural uncertainty lies in unmeasured confounders like customer satisfaction and service quality, which may significantly influence both contract choices and churn rates.

### 3. Identification

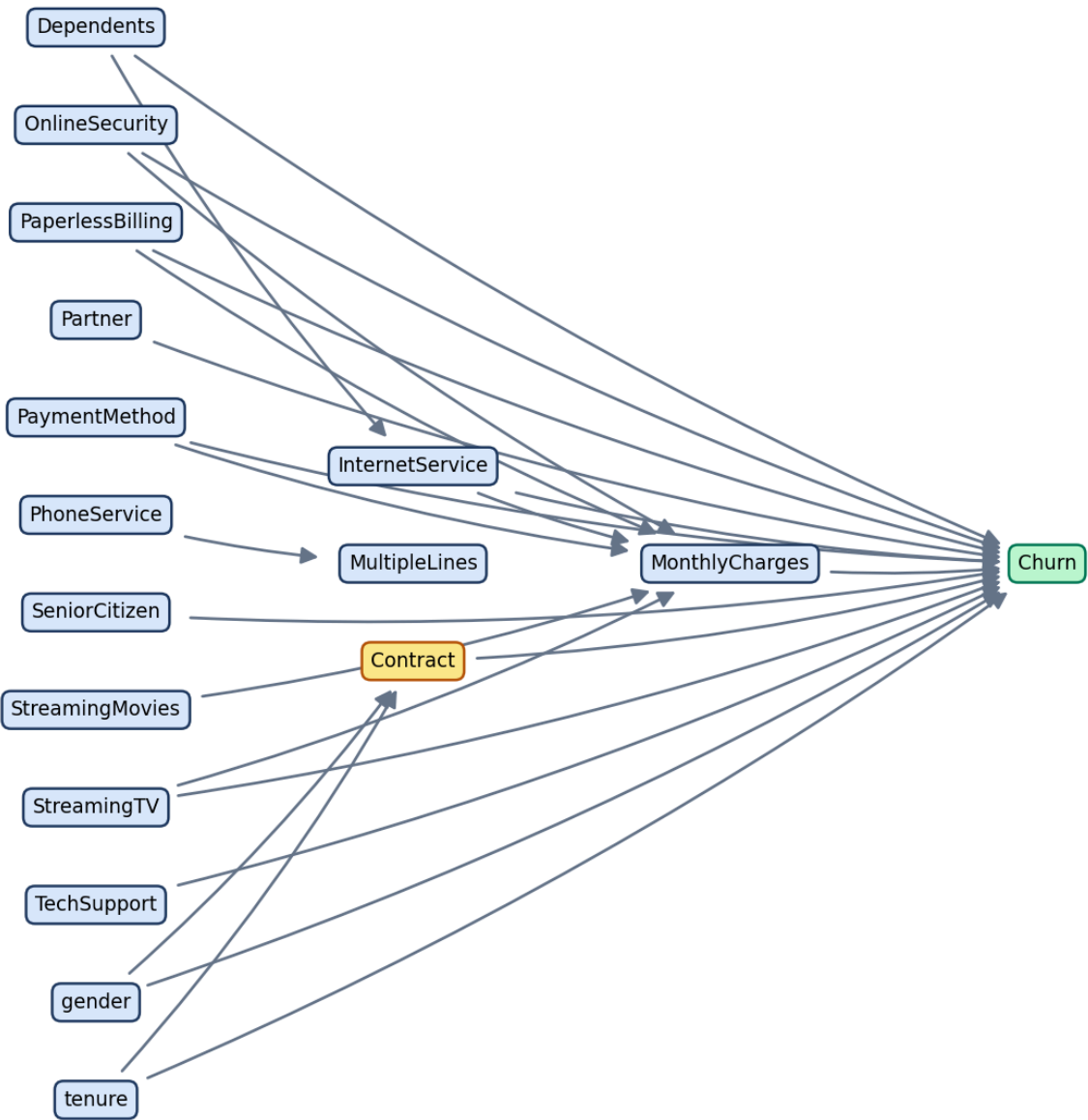
With tenure pointing into both Contract and Churn, the backdoor path  $\text{Contract} \leftarrow \text{tenure} \rightarrow \text{Churn}$  must be blocked; the second draft's (weakly supported)  $\text{gender} \rightarrow \text{Contract}$  edge adds gender. Minimal adjustment set: **{gender, tenure}** (badge: trust, n = 7,043). Dependents and InternetService influence services and charges but have no backdoor route to Churn through Contract, so they need no adjustment.

| Check               | Status | Detail                                          |
|---------------------|--------|-------------------------------------------------|
| Graph is a DAG      | pass   | Identification assumes an acyclic causal graph. |
| Treatment variation | pass   | Treatment must vary to support estimation.      |
| Outcome variation   | pass   | Low outcome variation weakens the analysis.     |

# Causal graph as saved to the canvas (full size)

Layered left-to-right by causal depth: sources on the left, the outcome on the right. Treatment highlighted in amber, outcome in green; arrowheads point from cause to effect.

treatment: Contract | outcome: Churn



| Check                | Status | Detail                                                     |
|----------------------|--------|------------------------------------------------------------|
| Backdoor path review | warn   | Open undirected paths may indicate unresolved confounding. |

## 4. The estimate: naive vs. adjusted vs. worst case

| Model                 | What it assumes                                                                                     | Effect per contract tier |
|-----------------------|-----------------------------------------------------------------------------------------------------|--------------------------|
| Naive (no adjustment) | Naive: only direct edges into Churn - equivalent to the raw comparison.                             | -0.2267                  |
| Adjusted (headline)   | The merged model: adjusts for gender, tenure.                                                       | -0.1455                  |
| Confounder-stressed   | Worst case: every direct cause of Churn also assumed to drive Contract, forcing maximal adjustment. | -0.0479                  |

The three numbers bracket the truth: -0.227 (naive) overstates because loyal customers pick long contracts *and* stay; -0.145 after blocking the tenure path; -0.048 if every churn driver also shaped contract choice. Direction is unanimous (79% magnitude spread, verdict: moderate).

## 5. Robustness

### Refutation tests

| Test                | Result                      | Interpretation                                                                                              |
|---------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------|
| placebo treatment   | status: ok, delta: 1.45e-01 | With a permuted fake treatment the effect collapses to 0 - the pipeline does not manufacture effects.       |
| dummy outcome       | status: ok, delta: 1.43e-01 | Against a synthetic outcome the effect vanishes (-0.003).                                                   |
| random common cause | status: ok, delta: 1.00e-05 | An irrelevant random covariate moves the estimate by 0.00001 - adjustment is stable.                        |
| data subsample      | status: ok, delta: 6.00e-04 | An 80% random subsample shifts the estimate by 0.0006 - the effect does not depend on particular customers. |

Passing a refutation test means the estimate is not a pipeline artefact and is stable to sampling; it cannot rule out an unmeasured confounder. That question is the job of the sensitivity analysis.

### Sensitivity to an unobserved confounder (Cinelli-Hazlett bias bound)

| Assumed unobserved-confounder strength (partial R <sup>2</sup> with treatment AND outcome) | Bias bound | Adjusted effect if such a confounder existed |
|--------------------------------------------------------------------------------------------|------------|----------------------------------------------|
| 0.1                                                                                        | 0.06861    | -0.07687                                     |
| 0.2                                                                                        | 0.1455     | 6.592e-05                                    |
| 0.3                                                                                        | 0.2334     | 0.08791                                      |
| 0.4                                                                                        | 0.3361     | 0.1906                                       |
| 0.5                                                                                        | 0.4603     | 0.3148                                       |

Robustness value: **0.21** - an unobserved confounder would need to explain 21% of the residual variance of *both* Contract and Churn to bring the effect to zero. Benchmark: tenure, the strongest measured confounder, explains 45% of the residual variance of Contract but only 1.6% of Churn given Contract. A hidden factor

thirteen times stronger than tenure on the churn side is implausible for anything a telco does not already record - which is why the effect is credible. (DoWhy's own partial-R<sup>2</sup> routine cannot run on this graph because it contains effect modifiers; the bound above is computed directly from the adjusted regression.)

## 6. Conclusions

- **Contract commitment reduces churn:** about 15 percentage points per tier after adjusting for tenure, against a 26.5% base rate.
- **The naive number is inflated:** reading the raw gap (-0.23) as causal would overstate the payoff of contract-upgrade campaigns by more than half.
- **The effect is robust:** all refutations pass, subsamples agree, the worst-case structure keeps 5 points, and the robustness value (0.21) is far above the strength of any measured confounder on the outcome side.
- **Actionable read:** incentives that move month-to-month customers onto annual contracts plausibly buy retention; a randomized encouragement design (offer the incentive to a random subset) would confirm it and separate lock-in from loyalty.

## Limitations (expanded after review)

- **Tenure is partly post-treatment.** A two-year contract mechanically generates tenure, so adjusting for snapshot tenure removes some of the contract's own effect; only tenure *at the time of the contract decision* is a clean confounder. The estimate is therefore conservative, not inflated.
- **Ordinal 0/1/2 coding assumes equal tier steps,** but the raw churn drop is 31 points from month-to-month to one-year and 9 points from one-year to two-year. Binary contrasts per tier would be more honest; the 0/1/2 ordering is also an alphabetical accident of the app's category encoding.
- **Linear probability model:** fitted churn probabilities are negative for about 16% of customers; a logistic model would give an odds-ratio effect instead. The app currently offers no GLM estimator.
- **Outcome definition:** "churned" includes customers who simply could not leave because they were mid-contract; the effect mixes lock-in with retention the company earns.
- **Cross-sectional prevalent-cohort snapshot:** customers who already churned before the window are absent (survivorship). A customer-month panel with within-customer contract changes is the telco-native design.
- **Only tenure and gender are adjusted for,** because that is what the drawn graph implies; acquisition channel, promotions and satisfaction are unmeasured here (see the variables section).

## Measuring an improved world model: which variables to add

The causal model above is only as good as the variables it contains. A better world model is not one with more columns, but one where every variable that opens a backdoor path between treatment and outcome is measured and adjusted for, where mediators are recognised and left alone when the total effect is the target, and where at least one source of quasi-random variation in the treatment exists to cross-check the observational estimate. The table lists the variables reviewers judged most valuable for this question, each with the causal role that decides how it must be used.

A world model for a causal question is the graph of who causes whom, and a new variable improves it only if it changes what the graph implies about the treatment-to-outcome effect. Ask two things about every candidate: when was it determined relative to the treatment decision, and does it sit on a path between treatment and outcome? A confounder is settled before the treatment and causes both the treatment and the outcome (governance sets the health budget and lowers mortality; service problems before the contract decision discourage commitment and cause churn); leaving it out leaves a backdoor open, so it belongs in the adjustment set. A mediator is caused by the treatment and carries part of its effect to the outcome (spending buys vaccine coverage; a term contract creates months remaining and an exit fee); adjusting for it throws away the part of the effect you are trying to measure, so measure it to understand the mechanism but keep it out of the adjustment set. A collider or outcome descendant is caused by the outcome, or by both the treatment and the outcome (life expectancy, Churn Score, a retention offer); adjusting for it manufactures an association that was never there. An instrument moves the treatment but reaches the outcome only through it (a randomised incentive offer); it is never adjusted for, and it lets you identify the effect without having measured every confounder.

The same column can play different roles depending on when it was measured, which is why the review kept asking 'measured when?'. Tenure at the contract decision is a confounder, while tenure at the snapshot is partly a mediator because the contract itself generates months of tenure. Satisfaction recorded before signing is a confounder, while the IBM snapshot satisfaction score is an outcome leak. HIV prevalence in the year 2000 is a confounder, while current prevalence is partly a descendant of the ART that health spending paid for. Health spending as a total is a bundle whose donor and out-of-pocket components are themselves caused by mortality, so redefining the treatment can matter more than any covariate. When you cannot pin down the timing (no contract start date, no survey timestamp), say so, state which way the resulting bias runs, and treat that ambiguity as the primary limitation rather than the last bullet.

Prioritise collection by the expected change in the answer per unit of effort, and remember that a bigger adjustment set is not automatically a safer one. First, use what is already in the file but absent from the graph: PaymentMethod and InternetService move the churn coefficient from -0.155 to -0.056, and a log transform with region dummies moves the World Bank t-statistic from -0.08 to -3.8, so free variables and free respecifications often move the estimate more than any hypothetical confounder sweep. Second, collect the pre-treatment version of the variable your identification already leans on (tenure at decision, domestic rather than total spending, lagged fertility). Third, add the named unobserved confounders that have confirmed, near-complete sources (governance, female education, WASH), choosing one variable per mechanism rather than several collinear proxies, and checking overlap so the model is not extrapolating across regions with no support. Fourth, add one mediator and one outcome refinement so the mechanism can be checked rather than assumed. Finally, plan the design change that identifies the effect without trusting the adjustment set at all, a randomised encouragement for churn or a within-country panel for the World Bank, and write down in advance which result would change your mind.

## Variables to collect, by causal role

| Variable                                                                                            | Causal role                                                                                                                                                                                                                                                             | Why it matters for this question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Where to get it                                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Tenure at the time of the contract decision (replaces snapshot tenure)</b>                       | Confounder - adjust for it (pre-treatment; snapshot tenure then becomes a partial mediator and leaves the adjustment set)                                                                                                                                               | Snapshot tenure equals tenure at the contract decision plus months survived under that contract, and the second part is caused by the contract (a two-year term mechanically extends survival, with 0% churn among the 68 two-year customers with tenure of 12 months or less), so adjusting for snapshot tenure conditions partly on a post-treatment quantity and the drop from -22.4pp to -14.2pp mixes genuine deconfounding with over-adjustment. Tenure measured when the contract was chosen carries exactly the tenure_pre -> Contract and tenure_pre -> Churn edges identification needs and is the single most important structural fix; keep snapshot tenure in the model until the pre-decision version exists, otherwise the backdoor reopens. | Derived from the operator's CRM or billing subscription table: tenure_pre = tenure minus months elapsed since contract_start_date, with term_end_date and renewal history from the same pull;                                  |
| <b>Customer satisfaction (NPS or CSAT) measured before the contract decision</b>                    | Confounder - adjust for it (pre-decision measurement only; a post-signing score is a mediator and the IBM snapshot score is an outcome leak)                                                                                                                            | Satisfied customers both commit to longer terms and stay regardless of contract, so satisfaction opens the backdoor Contract <- satisfaction -> Churn that tenure does not block, and the agent's own draft names it as the biggest structural uncertainty. Relationship NPS covers a small, non-random fraction of the base and post-interaction CSAT exists only for customers who contacted support, so treat it as a validation adjustment on the surveyed subsample rather than as the primary adjustment set.                                                                                                                                                                                                                                         | Production survey platform (Medallia, Qualtrics, IVR post-call surveys) joined on customer ID with the survey date restricted to before contract_start_date.                                                                   |
| <b>Support tickets and network outages in the 90 days before the contract decision</b>              | Confounder - adjust for it (windowed to before the decision; later tickets are mediators)                                                                                                                                                                               | Service problems before the decision discourage commitment and raise churn (Tickets -> Contract, Tickets -> Churn), giving an objective, full-coverage counterpart to the survey score for the 'service quality' confounder the report names, and service-call volume is one of the strongest known churn predictors in telecom. It is not the TechSupport column, which records a subscribed add-on rather than a contact history.                                                                                                                                                                                                                                                                                                                         | Ticketing system export (Zendesk, ServiceNow, Salesforce Service Cloud) with ticket timestamps and contact-centre ACD call logs, plus network outage logs, joined on account and windowed against the CRM contract start date; |
| Variable                                                                                            | Causal role                                                                                                                                                                                                                                                             | Why it matters for this question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Where to get it                                                                                                                                                                                                                |
| <b>Credit and payment-risk trait: PaymentMethod and autopay now, credit tier at onboarding next</b> | proxy confounder now (PaymentMethod and Contract are co-chosen at signup and jointly reflect an unmeasured credit trait); direct confounder once the onboarding credit band is available; autopay is a mediator if it was enrolled as a promo condition of the contract | Electronic-check customers are 47.7% of month-to-month versus 9.9% of two-year customers and churn at 45.3% versus 15-19% for other methods (autopay customers churn at 16.0% versus 34.7% and are 37% two-year versus 14%), yet PaymentMethod is absent from the merged graph, and adding it to the tenure-adjusted model moves the contract coefficient from -0.155 to -0.112. Credit decisioning gates postpaid term contracts and predicts involuntary non-payment churn, so the onboarding credit band is the clean pre-treatment version of the same trait; because snapshot payment method can change after signup and autopay may be a promo condition, estimate with and without it and report the range.                                          | Now: Churn.csv column 'PaymentMethod' (verified); autopay = PaymentMethod in {'Bank transfer (automatic)', 'Credit card (automatic)'} (verified, 3,066 of 7,043 rows).                                                         |



| Variable                                                                                    | Causal role                                                                                                                                                                                                                                         | Why it matters for this question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Where to get it                                                                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                             | (PaymentMethod and Contract are co-chosen at signup and jointly reflect an unmeasured credit trait); direct confounder once the onboarding credit band is available; autopay is a mediator if it was enrolled as a promo condition of the contract) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                           |
| <b>InternetService (fiber optic versus DSL versus none)</b>                                 | Confounder - adjust for it (pre-treatment product choice; parent of MonthlyCharges)                                                                                                                                                                 | Fiber customers are 54.9% of month-to-month versus 25.3% of two-year customers and churn at 41.9% versus 19.0% for DSL and 7.4% for no internet, and product line is selected at signup and drives contract type (fiber promotions), price and churn; adding this one observed column to the tenure-adjusted model moves the contract coefficient from -0.155 to -0.069 (to -0.056 together with PaymentMethod), more than the report's entire confounder-stress sweep. Because it is the parent of MonthlyCharges, adjust for InternetService rather than for MonthlyCharges, which is plausibly a mediator of Contract -> Churn, and one-hot encode it rather than using the {DSL: 0, Fiber optic: 1, No: 2} codes. | Churn.csv column 'InternetService' (repository root; verified present).                                                                                                                                                                   |
| <b>Signup sales context: acquisition channel and promotional offer at signup or renewal</b> | Confounder - adjust for it (only when the offer precedes the contract decision; a post-decision retention offer is post-treatment)                                                                                                                  | Retail stores and dealers push 24-month bundles while web self-serve skews month-to-month, dealer- and door-to-door-acquired customers churn more for reasons unrelated to the contract, and promotional offers move price-sensitive customers onto term contracts who leave when the discount expires, so channel and offer add Channel/Offer -> Contract and Channel/Offer -> Churn and are the most practical proxies for the 'price sensitivity' the report lists as unmeasured. Both are fixed on the activation or renewal order and are natural stratification variables for the incentive experiment; the promo attached to the new contract itself is part of the treatment bundle, not a confounder.        | Offer, obtainable now: IBM Cognos Analytics 'Telco customer churn (11.1.3+)' sample, Telco_customer_churn_services.xlsx column 'Offer' (None / Offer A-E), joinable on Customer ID to the same 7,043 customers (Kaggle mirrors ylchang... |
| <b>Competitor availability at the service address</b>                                       | Confounder - adjust for it (exogenous, area-level); effect modification as a secondary CATE breakdown)                                                                                                                                              | Where there is no alternative provider month-to-month customers do not churn either, so competition changes both contract take-up (Competition -> Contract) and churn (Competition -> Churn), which turns the 'competitor offers' limitation into a measurable covariate and a geographic fixed-effect proxy, and competitor entry dated by area is additionally a candidate difference-in-differences shock. It is the weakest of the churn confounders and, because the IBM customers are synthetic, a join to real competition data will likely show nothing, so present it as a worked example of an external join.                                                                                               | FCC Broadband Data Collection / National Broadband Map (broadbandmap.fcc.gov/data-download): location- and census-block-level fixed availability with provider counts, needing a block-to-ZIP crosswalk because the Area Summaries sto... |

| Variable                                                                                                                                   | Causal role                                                                                                                               | Why it matters for this question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Where to get it                                                                                                                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Months remaining on term, early-termination fee exposure and the promo attached to the contract (from contract start and end dates)</b> | Mediator - do NOT adjust (total effect) (never adjust for the total effect; use for phase-specific effects and the term-expiry hazard)    | Churn hazard for term customers concentrates in the renewal window and the remaining fee is the mechanical lock-in channel (Contract -> months_remaining/ETF -> Churn), while a promo bundled with the new contract sits on Contract -> Promo -> MonthlyCharges -> Churn, so all three are descendants of the treatment and adjusting for them blocks the effect. Measuring them lets the example separate 'contracts retain because leaving costs money' from 'committed customers are happier' through the churn spike at term expiry and supports a mediation decomposition; front-door identification is not credible because the fee is not the sole channel and is near-collinear with tier times months remaining. | Billing contract terms table:<br>contract_start_date and term_end_date<br>(months_remaining = end date minus observation month, 0/NA for month-to-month), the ETF schedule and remaining amount, and the promo-code table with discount amou... |
| <b>Randomized contract-incentive offer exposure (encouragement design)</b>                                                                 | Instrument - enables an IV design (valid only if a randomized test or holdout was actually run and exposure, not acceptance, is recorded) | Randomly assigned encouragement to sign a 12- or 24-month term shifts Contract by design and, if the incentive is a one-off non-price benefit, affects churn only through the contract, yielding an intention-to-treat effect of the campaign and a complier (LATE) effect of contract signing; it is the only proposal that identifies the effect without measuring every confounder and it is the A/B test the README already recommends. If the offer bundles a persistent discount the IV identifies the contract-plus-discount bundle, which should then be the stated estimand, and retention offers targeted at high-risk accounts violate independence.                                                           | Campaign-management or experimentation-platform test/control cell assignment log keyed on customer ID and offer date, recording who was shown the offer; the IBM 'Offer' column records accepted offers and cannot stand in.                    |
| <b>Contract as two binary contrasts (month-to-month versus one-year; one-year versus two-year)</b>                                         | Treatment refinement (not an adjustment variable)                                                                                         | DoWhy evaluates control_value 0 versus treatment_value 1 with interaction terms rather than a per-tier slope, and the raw churn rates of 42.7% / 11.3% / 2.8% show the two steps are far from equal (tenure-adjusted -25.3pp and -29.9pp versus month-to-month, so the second step is about -4.7pp), so the single coefficient is mislabelled. Binary contrasts also make the treatment binary, which unlocks the propensity-score and doubly robust estimators, overlap diagnostics and the partial-R2 sensitivity that the three-level coding currently rules out.                                                                                                                                                      | Derived from Churn.csv 'Contract': an indicator for rows with Contract in {Month-to-month, One year} and separately for rows in {One year, Two year}; no new data needed. Proposed by three lenses.                                             |
| <b>Churn reason, voluntary versus involuntary churn, and fixed-horizon survival or renewal at term end</b>                                 | Outcome refinement (placebo) outcome (never adjust)                                                                                       | Contracts can plausibly act only on voluntary churn (competitor, price), not on moves, deaths or the non-payment disconnects that autopay governs, so splitting the outcome isolates the mechanism the business can act on and gives a placebo outcome that a genuine lock-in effect should not move, while survival to a fixed horizon after the decision and renewal when the customer is first free to leave replace 'churned this month' with retention the operator actually earns rather than churn postponed to the cliff. Reasons exist only for churners, so build per-customer indicators, and the 'Moved' and 'Deceased' placebo is low-powered.                                                               | IBM Cognos Telco_customer_churn_status.xlsx columns 'Churn Category' (Attitude, Competitor, Dissatisfaction, Other, Price;                                                                                                                      |
| Variable                                                                                                                                   | Causal role                                                                                                                               | Why it matters for this question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Where to get it                                                                                                                                                                                                                                 |
| <b>Post-treatment and post-outcome columns:</b>                                                                                            | Collider - never adjust (never adjust; the                                                                                                | TotalCharges is approximately tenure times MonthlyCharges (a descendant of the price channel and of tenure recorded at churn), Churn Score and                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Churn.csv<br>'TotalCharges'; IBM Telco_customer_churn_st                                                                                                                                                                                        |

| Variable                                                                                                                       | Causal role                                               | Why it matters for this question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Where to get it                                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>TotalCharges, Churn Score, CLTV, Number of Referrals, retention offer received, and the snapshot survivorship selection</b> | survivorship node is a design limitation, not a variable) | CLTV are model predictions trained on churn labels, Number of Referrals follows from a satisfied ongoing relationship, and save-desk offers are triggered both by approaching term end and by expressed dissatisfaction (Contract -> Offer <- Dissatisfaction -> Churn), so conditioning on any of them either opens a path through the outcome or blocks the effect. Presence in the cross-section itself requires surviving to the snapshot month, which both the contract and unobserved loyalty cause, so the dataset is already conditioned on a selection collider that only an activation-cohort extract retaining churners can undo. | atus.xlsx 'Churn Score' and 'CLTV'; Telco_cust omer_churn_services.xlsx 'Referred a Friend' and 'Number of Referrals' (all verified); |

Sources are abbreviated here; the README in this folder carries the full provenance notes (indicator codes, verified endpoints, joins) for every variable

Reading the role column: a confounder belongs in the adjustment set; a mediator must be left out when estimating the total effect (adjusting for it removes part of the very effect being measured); an instrument enables an alternative identification strategy; a collider must never be adjusted for. Collecting a variable is half the work - placing it correctly in the DAG is the other half.

## Other ways to strengthen this example (reviewer findings, verified)

Fifteen improvements survived adversarial verification across both examples; those relevant here are listed with their status after this revision. Items marked open are the roadmap.

- Replace the fixed-formula confounder-strength sweep with a real sensitivity analysis and delete the false churn-versus-World-Bank contrast** - *DONE in this edition; small effort, high value.* In services.py summarize\_robustness (lines 863-874) the sweep is `adjusted_effect = baseline_estimate * (1 - 0.35 * strength)` with no data input at all, so replace it with DoWhy's `add_unobserved_common_cause` run over a grid of `effect_strength_on_treatment` by `effect_strength_on_outcome` calibrated against observed covariates, and compute the Cinelli-Hazlett robustness value directly from the regression t-statistic (RV = 0.21 for the tenure-adjusted churn LPM with  $t = -19.9$  and  $df = 7040$ , versus a partial  $R^2$  of tenure with Churn of only 0.016; RV = 0.0058 for the World Bank levels model), plus an E-value for the binary churn outcome. Where DoWhy refuses the partial- $R^2$  analysis because of effect modifiers, run it on the binary contrasts or compute RV from the fitted regression outside DoWhy instead of papering over the gap with the cosmetic table. In the documents, delete the churn PDF sentence claiming 'the same sweep drove the [World Bank] effect essentially to zero', remove the churn README bullet that the effect 'barely moves under the unobserved-confounder sensitivity sweep', either drop the sweep table or relabel it as an illustrative linear shrink, and add a short box listing which robustness numbers actually carry information (robustness value, t-statistic, competing-model spread) and which are sanity checks.
- Report the churn treatment as two binary contrasts instead of a single per-tier slope** - *open; small effort, high value.* Recode Contract into month-to-month versus one-year (rows with Contract in the first two levels) and one-year versus two-year (rows in the last two), or any-commitment versus month-to-month, and report the raw churn rates (42.7% / 11.3% / 2.8%) next to the tenure-adjusted contrasts (-25.3pp for one-year and -29.9pp for two-year relative to month-to-month, so the one-year to two-year step is only about -4.7pp). Add conditional effects by tenure bin (the contract by tenure interaction gives -30pp at 3 months and -8.7pp at 60 months) and overlap diagnostics (only 29 two-year

customers with tenure of six months or less versus 1,413 month-to-month; 1,263 versus 342 above 48 months). Have the agent auto-propose binarisation whenever an ordinal treatment has five or fewer levels, because `recommend_estimator` (`agent_service.py` lines 653-715) gates propensity-score, weighting and doubly robust estimators and the partial-R2 sensitivity on a binary treatment and `heuristic_role_candidates` (lines 546-557) requires a binary column. State that the 0/1/2 ordering is an alphabetical accident of category-code encoding and that DoWhy evaluates `control_value` 0 versus `treatment_value` 1 with interaction terms rather than an average per tier.

- State the tenure-versus-contract time ambiguity and contractual lock-in as the churn example's primary limitation and reframe the estimand** - *PARTLY done; small effort, high value*. Add a paragraph and a small two-panel DAG to the churn README and PDF ('what we drew' with tenure -> Contract next to 'what may also be true' with Contract -> tenure) and give the numbers: mean tenure is 18.0 / 42.0 / 56.7 months by tier, the tenure-contract correlation is 0.67, 51.5% of month-to-month customers versus 4.0% of two-year customers have tenure of 12 months or less, and the 68 two-year customers with tenure of 12 or less (158 with tenure of 24 or less) show exactly 0% churn, which is lock-in mechanics rather than loyalty. State the estimand honestly as the effect conditional on having survived to the current tenure, under the assumption that the contract decision preceded the accumulation of tenure, and say that Churn.csv cannot verify this because it has no contract start or renewal date; phrase the bias as 'attenuated toward zero if tenure is partly a mediator; unbiased only if tenure is purely pre-treatment' rather than calling -14.2pp an upper bound. Add an outcome-window note (Churn means left within the observation month; part of the effect is the mechanical impossibility of leaving mid-term without a fee; term-end renewal is unobserved; the cross-section is a length-biased prevalent cohort), promote this from the last bullet to the first limitation, and drop the 'genuine, actionable lever' wording. Longer term, express the estimand as a discrete-time hazard (a logit with tenure as the clock and contract as a covariate) or restrict to a signup cohort with contract chosen at activation, and name tenure-at-decision as the variable that resolves the ambiguity.
- One consistency pass: a shared glossary, percentage points everywhere, per-refuter delta semantics, and the stale or contradictory statements in both READMEs and PDFs** - *PARTLY done; small effort, high value*. Add a shared glossary covering backdoor path, adjustment set, naive versus adjusted, refuter, what delta means for each refuter type, refuter p-value, robustness value and partial R2, robustness score, effect modifier in DoWhy's sense, linear probability model, percentage point, elasticity, ecological data, and the minimal and confounder-stressed variants. Fix the churn PDF's '14.2%' (page 1) and '5.7% of the effect remains' (page 4) to percentage points and note that the relative reduction against the 27% base rate is roughly 53%; explain that placebo and dummy-outcome refuters pass with a delta equal to the effect while random-common-cause and subsample refuters pass with tiny deltas, and correct churn README line 53 ('tiny deltas'); explain the 'Backdoor path review: warn' line in the churn PDF, not just in the World Bank footnote; correct World Bank README step 6 (refuters test pipeline sanity and do not discriminate the naive from the adjusted model); remove the dangling 'use-case design' reference at World Bank README line 5; present region as a confounder rather than an 'optional grouping variable'; reconcile the World Bank page 6 note that the partial-R2 analysis was 'unavailable in section 5' with page 5 reporting a robustness value of 0.0058 and state that the section 4 numbers were regenerated after the `compare_models` fix; remove 'tenure's direct edge' from the churn page 5 list of effect modifiers (DoWhy's set is SeniorCitizen, PaperlessBilling, MonthlyCharges, StreamingTV, InternetService and TechSupport, and tenure is the backdoor variable); fix the root README's stale CausalGUI paths (lines 158 and 167) and the './README.md' link; and stop describing Churn.csv as a built-in dataset when the walkthrough starts with an upload.

- Make refutation results informative: seeds, real p-values, pass criteria, null-baseline labelling, more simulations, and refuters that can actually discriminate** - *open; small effort, high value*. In `services.py` accept an optional `random_state` on `/api/robustness_dashboard/` and pass it to the four refuters (lines 748-756 currently pass none, so deltas differ every run and the World Bank subsample flips sign between seeds); read p-values from `result.refutation_result['p_value']` instead of `getattr(result, 'p_value')` (lines 674-692), which is always None in DoWhy 0.14; replace the unconditional status 'ok' at line 687 with per-refuter pass/fail criteria based on a p-value threshold and sign consistency; raise the `data_subset num_simulations` from 5 to at least 100 (line 743); label refuter results 'not informative' whenever  $|t|$  of the baseline is below about 1 and exclude them from the score; and make `summarize_robustness` (lines 844-883) stop scoring the fraction of checks that completed. Add refuters that discriminate: bootstrap confidence intervals, leave-one-country-out and leave-one-region-out for the 186-row cross-section, a negative-control outcome (gender or PhoneService as the outcome for Contract), a functional-form refuter (log versus level), and an 'all valid adjustment sets agree' check, which is cheap because identification already enumerates the sets. In the reports print the seed and p-values in every refutation row, replace 'all four passed' and 'robustness score 100%' with statements backed by p-values, and drop the World Bank 'never the reverse' direction claim, which reads the sign of a coefficient with  $t = -0.08$ .
- Add the observed pre-treatment confounders to the churn DAG and fix how the competing model variants are constructed** - *PARTLY done; medium effort, high value*. Add `PaymentMethod` (or an autopay indicator) and `InternetService` as confounders in the example graph: electronic check is 47.7% of month-to-month versus 9.9% of two-year customers and churns at 45.3% versus 15-19%, and fiber is 54.9% versus 25.3% and churns at 41.9% versus 7.4% for customers without internet. Report the measured-covariate bracket alongside the sweep: tenure only -0.155 (ordinal), plus `InternetService` -0.069, plus `PaymentMethod` -0.056, all observed pre-treatment covariates -0.032 ( $t = -3.8$ ), and one-hot all-covariate contrasts of -0.106 (one-year) and -0.063 (two-year). In `build_model_variants` (`agent_service.py` lines 886-947) generate a 'full observed pre-treatment adjustment' variant from the dataset columns rather than the canvas edges (`compare_model_variants` already receives the data frame), let the analyst tag each column as pre- or post-treatment so mediators are excluded, exclude the naive 'minimal' variant from the stability verdict because it deletes every edge into the treatment and is the naive estimate by construction, and soften the 'worst-case confounding scenario' description string (lines 927-929). Rewrite the README and PDF sentence about -5.7pp as 'a variant that additionally adjusts for price and service variables gives -5.7pp; because `MonthlyCharges` (66.4 / 65.1 / 60.8 by tier) and `TechSupport` are plausibly downstream of `Contract`, this is a different estimand, not a bound', and explain which observed variables were deliberately left unadjusted and why.
- Report standard errors and confidence intervals for every estimate and add a logit or GLM estimator whenever the outcome is binary** - *open; medium effort, high value*. Add `backdoor.generalized_linear_model` with a Binomial family and average marginal effects to `recommend_estimator` (`agent_service.py` lines 653-715, which today only ever returns linear\_regression or propensity methods) and to the Robustness Dashboard's estimator comparison whenever the outcome is binary, so LPM and logit appear side by side. Use DoWhy's `get_standard_error` and `get_confidence_intervals`, or bootstrap or robust standard errors, for every reported effect, and put estimate, standard error, 95% CI and n in every row of the model-comparison and refutation tables in the API payload and in both example documents (there is no report template in the repo, so the PDF change is manual). Quote the diagnostics: the tenure-adjusted LPM produces fitted probabilities outside [0, 1] for 1,140 of 7,043 customers (16%), the logit average marginal effect for the ordinal specification is

about -21.5pp versus -15.5pp from OLS, and the binary-contrast logit AMEs should be re-derived before quoting because reviewers obtained -0.24/-0.33 rather than the -0.22/-0.42 first reported.

- **Ship results.json and graph.json for each example, add a verify script, and document the exact DoWhy specification behind every number** - *open; medium effort, high value*. Add examples//graph.json (the edge list in the shape /api/save\_graph/ accepts) and examples//results.json (dataset sha256, dowhy and pandas versions, cleaning steps applied, edges with verifier verdicts, treatment and outcome, adjustment set, method and contrast, per-variant estimates, refutations with p-values and seed, partial-R2 output), plus a verify\_examples.py or Django test that loads the CSV, applies the cleaning steps, calls estimate\_effect with graph.json and asserts the estimates within 1e-6 (the review replay reproduced -0.141775, -0.223685, -0.057381 and -1.54434e-05 exactly in about 40 lines of existing functions). In each report state the estimator, the control-0-versus-treatment-1 contrast, the adjustment set and the effect-modifier set as DoWhy resolved them, rename the 'naive / no adjustment' rows to 'minimal direct-effects model with effect-modifier interactions', and give the actual raw comparisons (the raw churn slope is -0.210 and the raw World Bank slope is -8.23e-04, so the '104x' ratio is about 53x and the '58% overstatement' is 36%). Make the README's re-run-with-context step reproducible by adding a context textbox to the Causality Agent panel or documenting the exact HTTP call (AppRoot.vue line 1015 hard-codes the context while views.py line 1130 accepts it), and record OPENAI\_MODEL, date and verbatim prompts in results.json with a note that the LLM step is not byte-reproducible. Add examples/churn/DATA.md with source (IBM Telco sample via Kaggle blastchar), license, sha256, shape 7,043 x 21, the 11 blank TotalCharges rows and a column table listing the category codes the app assigns.
- **Give both READMEs the same skeleton with a headline block, a 'what would change my mind' box, a next-steps checklist mapped to app stages, and a variables-to-collect table** - *PARTLY done; medium effort, high value*. Use one skeleton for both READMEs (Question, Data, DAG, Adjustment set, Naive versus adjusted with units, Robustness evidence that carries information, What would change my mind, Next data to collect, Reproduce in the app, Glossary link) and add the missing headline block to the World Bank README. Write the boxes: for churn, tenure measured at contract start shrinking the effect, the effect concentrating entirely in the month-to-month to one-year step, an encouragement experiment giving a near-zero complier effect, and adjusting for the acquisition offer removing the effect; for the World Bank, a within-country fixed-effects estimate on the panel (labelled outside-the-app, because Time Series mode is rolling-window correlation, not a fixed-effects estimator), government spending share instead of total spending, and neonatal mortality responding more than post-neonatal, while dropping 'a significant log-log elasticity, which already holds' because that is handled as a functional-form fix in item 6. Add the encouragement design to the churn next steps with the intention-to-treat versus complier distinction: randomise a one-off non-price incentive at renewal or at month 3-6 of month-to-month tenure, stratify on tenure-at-decision, channel and autopay, pre-register 12-month survival and gross margin, expect roughly 1,500 customers per arm to detect a 5pp reduction at 80% power, and analyse the offer as an instrument through the app's iv.instrumental\_variable path (services.py line 158). Say explicitly that binary contrasts and log specifications currently require pre-transforming the CSV before upload, and populate the 'Variables to collect next' table from the variable tables in this synthesis.
- **Redraw the DAG figures with arrowheads and full labels and add Mermaid DAGs to both READMEs** - *open; medium effort, high value*. generate\_graph\_image uses a spring layout with node\_size 800 and arrowsize 10, so arrowheads sit under the node discs and World Bank labels clip to

'hild\_mortality\_rate' and 'health\_expenditure\_per\_'; switch to a layered left-to-right Graphviz dot layout with visible arrowheads, full labels and distinct styling for treatment, outcome and adjustment nodes, and regenerate the page 3 figures in both PDFs. Add a Mermaid DAG block to each README (GitHub renders these natively) with the adjustment set highlighted and the backdoor path labelled; in the churn README show the 'star' first draft next to the merged DAG so walkthrough step 4 becomes visual; and add a two-panel naive-versus-adjusted bar with units to the root README's Example Datasets section.

- **One-hot encode nominal categoricals, add a nominal-as-ordinal profiler check, and teach the heuristic role detector domain vocabulary** - *open; medium effort, medium value*. In `preprocess_data_frame_for_causal` (`services.py` lines 103-106) keep nominal text columns as category dtype or one-hot encode them (drop-first) so DoWhy's regression estimators dummy them, pass the dummy set to DoWhy as the common cause, and keep an ordinal path only for columns the analyst explicitly marks as ordered (Contract is one); add a profiler issue type such as 'nominal\_categorical\_as\_ordinal' (`profile_data_frame` lines 151-240 has none) and expose the encoding choice as a reviewable cleaning step; and stop the edge verifier from running Pearson correlations on category codes, because InternetService becomes {DSL: 0, Fiber optic: 1, No: 2} against mean MonthlyCharges of 58 / 92 / 21 and TechSupport becomes {No: 0, No internet service: 1, Yes: 2} against churn of 41.6% / 7.4% / 15.2%, yet the churn PDF reports those edges as 'supported, 1.00'. Add 'expenditure', 'spending' and 'coverage' to `TREATMENT_NAME_PATTERN` and 'mortality', 'death', 'rate', 'incidence' and 'prevalence' to `OUTCOME_NAME_PATTERN` (`agent_service.py` lines 30-36) and allow non-binary treatment candidates on a name hit (lines 546-559) so the fallback can propose roles for the World Bank dataset without an API key.

## What changed in this edition

A multi-lens review of the first edition found that the app's "confounder-strength sweep" was a fixed arithmetic formula presented as evidence; it has been replaced by the Cinelli-Hazlett bias bound, which - unlike the old sweep - actually bounds unmeasured confounding and can be benchmarked against observed variables. The limitations section was expanded with the reviewers' verified points (tenure as partly post-treatment, unequal contract tiers, the linear probability model's out-of-range fits, and the lock-in vs. loyalty distinction). The analysis itself was re-run; the headline effect moved from -0.142 to -0.145 because the merged graph now also adjusts for gender.